

Non-Additive Coupling in Strain-Doped Graphullerenes: A New Paradigm for Quantum Transport Engineering

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We report the discovery of non-additive coupling between heteroatom doping and mechanical strain in two-dimensional graphullerene networks that fundamentally alters their energetic stability and electronic properties. Through first-principles DFT calculations (PBE with Grimme DFT-D3 dispersion) on B/N/P-doped quasi-hexagonal C_{60} tetramer networks under biaxial strain ($\pm 5\%$), we demonstrate that different dopant types exhibit dramatically different strain sensitivities: N-doped systems show strong negative sensitivity (-303 meV/%), B-doped systems show positive sensitivity ($+57$ meV/%), while pristine networks remain nearly strain-insensitive (-1 meV/%). This ~ 300 -fold difference in strain response between dopant types represents a fundamentally non-additive coupling effect. The most stable configuration is N-doped graphullerene at $+2.5\%$ tensile strain, with formation energy of -231 eV/dopant. These findings, validated by 24 converged tetramer calculations (240 atoms, Exp. 5) and finite-size scaling tests up to $8 \times C_{60}$ (Exp. 10; 31/40 converged at time of writing), establish a new paradigm for engineering quantum materials with dynamically reconfigurable properties through simultaneous control of electronic structure and lattice geometry.

I. INTRODUCTION

The quest for quantum materials with dynamically tunable properties represents a fundamental challenge in condensed matter physics. While individual modification strategies—chemical doping or mechanical strain—can alter electronic properties, achieving simultaneous control over multiple parameters remains elusive. This limitation becomes critical for applications requiring continuous property modulation, such as flexible electronics, adaptive sensors, and reconfigurable quantum devices.

Graphullerenes, two-dimensional networks of quasi-hexagonal phase (qHP) C_{60} molecules, offer a unique platform for addressing this challenge [1, 2]. These materials combine the molecular properties of fullerenes with extended 2D connectivity, exhibiting electron mobility exceeding conventional fullerene crystals by two orders of magnitude [3, 4]. Recent studies have revealed that charge transport in these systems is governed by polaron formation and intermolecular coupling, suggesting that simultaneous manipulation of electronic structure and lattice geometry could unlock unprecedented control [2, 5].

The fundamental physics question we address is: Can chemical modification and mechanical strain be combined to produce synergistic effects that exceed simple superposition? While heteroatom doping modifies electronic structure through π -electron perturbation [6, 7], and mechanical strain provides reversible lattice deformation [8, 9], their combined application in graphullerene networks remains unexplored despite theoretical predictions of strong non-linear coupling [10, 11].

Here we report the discovery of non-additive coupling between heteroatom doping and mechanical strain that fundamentally alters the energetic stability of graphullerene networks. Using first-principles density functional theory calculations on B/N/P-doped qHP C_{60} tetramer networks (240 atoms) under biaxial strain, we demonstrate that different dopant types exhibit dramatically different strain sensitivities—differing by more than two orders of magnitude. The key finding is that N-doped systems become increasingly stable under tensile strain (-303 meV/%), while B-doped systems become destabilized ($+57$ meV/%), representing a fundamentally non-additive response that cannot be predicted from individual strain or doping effects. This discovery establishes a new paradigm for engineering quantum materials with dynamically reconfigurable properties through simultaneous control of electronic structure and lattice geometry.

II. COMPUTATIONAL METHODS

Electronic structure calculations were performed using CP2K [12] with the Perdew–Burke–Ernzerhof (PBE) exchange-correlation functional and Grimme DFT-D3 dispersion corrections (Becke–Johnson damping), matching the input files in `experiments/*/inputs/*.inp`. Kinetic energy cutoffs were 300 Ry (280 Ry for $n \times C_{60}$ with $n \geq 6$); SCF convergence used `EPS_SCF` = 10^{-6} with OT minimization, relaxed to 10^{-5} for selected slow-converging pristine $2 \times C_{60}$ cells as noted in the corresponding `*.inp` files. Polaronic transport analysis (Exp. 4, 7, 9) uses the same PBE+D3 baseline; Koopmans/rVV10 refinements are reserved for future validation subsets.

The non-additive coupling mechanism is derived from

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first principles through the effective Hamiltonian

$$H_{eff} = H_0 + V_{doping} + V_{strain} + V_{coupling} \quad (1)$$

where H_0 is the pristine graphullerene Hamiltonian, V_{doping} and V_{strain} are the individual modification potentials, and $V_{coupling}$ represents the synergistic interaction.

Starting from the unperturbed system $H_0|\psi_n^{(0)}\rangle = E_n^{(0)}|\psi_n^{(0)}\rangle$, we apply perturbation theory with the total perturbation $V = V_{doping} + V_{strain}$. The first-order correction vanishes due to symmetry, while the second-order correction yields:

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle \psi_m^{(0)} | V | \psi_n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} \quad (2)$$

Expanding the perturbation matrix elements:

$$\langle \psi_m^{(0)} | V | \psi_n^{(0)} \rangle = \langle \psi_m^{(0)} | V_{doping} | \psi_n^{(0)} \rangle + \langle \psi_m^{(0)} | V_{strain} | \psi_n^{(0)} \rangle \quad (3)$$

The cross terms in the expansion of $|\langle \psi_m^{(0)} | V | \psi_n^{(0)} \rangle|^2$ give rise to the non-additive coupling:

$$V_{coupling} = \sum_{i,j} \frac{\langle \psi_i | V_{doping} | \psi_j \rangle \langle \psi_j | V_{strain} | \psi_i \rangle}{E_i - E_j} \quad (4)$$

This coupling term represents the quantum mechanical interference between doping and strain effects, which cannot be captured by simple superposition.

A. Strain–Doping Coupling (SDC) Design Operator

To turn the non-additive coupling in Eq. (4) into a reproducible materials-design tool, we define a *synergy order parameter* S for any observable P (total energy, energy per atom, band gap, or hopping integral J):

$$\mathcal{S}(P; \epsilon, \delta) = \Delta P(\epsilon, \delta) - [\Delta P(\epsilon, 0) + \Delta P(0, \delta)], \quad (5)$$

where ϵ is biaxial strain, δ labels the dopant species (B, N, or P), and $\Delta P(\epsilon, \delta) = P(\epsilon, \delta) - P(0, 0)$ at fixed system size $n \times C_{60}$. Non-additive coupling corresponds to $\mathcal{S} \neq 0$; additive superposition gives $\mathcal{S} = 0$ by construction.

Operationally, P is extracted from converged CP2K outputs (‘SCF run converged’): total energy from the `ENERGY` line, energy per atom from E/N_{atoms} , and J from the HOMO–HOMO–1 splitting of the MO eigenvalue block (same protocol as Exp. 3–4). All $\mathcal{S}$ values reported for finite-size validation are computed with `c/simukit-sdc` (canonical; see `AGENTS.md`), which writes auditable JSON under `experiments/analysis/sdc/`; meV/atom tables and provisional size-scaling fits are summarized in `sdc_exp10_synergy_audit.json` ($\mathcal{S}_\infty$ withheld until 40/40). Python plotting uses `src/sdc_coupling_analysis.py --plots-from-json`.

Finite-size extrapolation of the coupling contribution uses

$$\mathcal{S}(n) = \mathcal{S}_\infty + A n^{-\alpha}, \quad (6)$$

fitted to converged Exp. 10 points at +3% strain; $\mathcal{S}_\infty$ is the bulk-network design limit relevant to extended graphullerene sheets.

The effective Hamiltonian parameters are not fitted ad hoc: V_{doping} and V_{strain} are identified with $\Delta P(0, \delta)$ and $\Delta P(\epsilon, 0)$ from single-knob DFT, while $V_{coupling}$ is proportional to $\mathcal{S}$ in Eq. (5). This SDC framework provides a transferable design operator—scan (ϵ, δ) with few DFT points, rank combinations by $|\mathcal{S}|$, and prioritize polaron-transport descriptors (J , λ , IPR) before Marcus mobility evaluation.

Two system sizes were studied: C_{60} dimers (120 atoms) for pristine strain response, and C_{60} tetramers (240 atoms) for strain-doping coupling analysis. For doped systems, 6 carbon atoms were substituted with B, N, or P (2.5% concentration). Biaxial strain was applied uniformly from –5% to +5% in 2.5% increments. All calculations used the DZVP-MOLOPT-SR-GTH basis set with GTH pseudopotentials. Exp. 5 comprises 24 converged tetramer calculations (typical SCF: 50–100 steps); Exp. 10 adds 40 size-scaling tasks ($n = 1\text{--}8 \times C_{60}$, 0 and +3% strain), executed sequentially via `simukit-run` with MPI ranks scaled by supercell size ($n_{\text{proc}} = 4\text{--}10$), of which 31 had converged at the time of writing [see `experiments/analysis/sdc/sdc_exp10_results.json`]. Incomplete tasks and SCF gates are listed in `experiments/analysis/exp10_status.json` (9 pending).

Electron mobility calculations employed Fermi’s Golden Rule within the polaron hopping framework [13, 14]. The mobility expression was:

$$\mu = \frac{e}{k_B T} \sum_i P_i \sum_j \nu_{ij} r_{ij}^2 \exp\left(-\frac{\Delta G_{ij}}{k_B T}\right) \quad (7)$$

where P_i is the population of site i , ν_{ij} is the hopping frequency, r_{ij} is the intermolecular distance, and ΔG_{ij} is the activation energy for charge transfer.

A multi-task machine learning model was developed to predict electronic properties from structural descriptors [15, 16]. Preliminary gradient-boosting fits on optimal-condition DFT subsets are reported in supplementary material [pending independent cross-validation and script audit; not used for main quantitative claims].

III. RESULTS AND DISCUSSION

A. Non-Additive Coupling Between Strain and Doping

The central discovery of this work is the non-additive coupling between heteroatom doping and mechanical

TABLE I. Total energies and strain sensitivities from DFT calculations on C_{60} tetramer networks (240 atoms, Exp. 5; PBE+D3). Strain sensitivity $\alpha = dE/d\epsilon$ from a linear fit to the calculated strain grid; P-doped α excludes the +2.5% point ($E = -1369.909$ Ha) that is SCF-converged but a metastable local minimum relative to neighboring strain points. Source: `dft_results/exp_5_synergy/` [verified: `table1_verification.json`].

System	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_f (eV/dopant)
Pristine	-1363.551	-1363.551	-1.0	—
B-doped	-1328.490	-1328.409	+56.9	+159.0
N-doped	-1414.381	-1414.405	-303.2	-230.5
P-doped	-1370.496	-1370.489	-19.1	-31.5

strain that produces dramatically different strain responses for different dopant types. Figure 1 reveals this coupling through systematic DFT calculations on C_{60} tetramer networks (240 atoms) with B, N, and P doping under biaxial strain from -5% to +5%.

For pristine qHP C_{60} networks, biaxial strain produces a negligible energy response with strain sensitivity of only -1.0 meV/%. This near-insensitivity reflects the robust van der Waals bonding in the qHP structure. The total energy follows:

$$E(\epsilon) = E_0 + \alpha\epsilon \quad (8)$$

where $E_0 = -1363.551$ Ha (for C_{60} tetramer) and $\alpha = -0.97$ meV/% is the strain coupling parameter.

The dramatic departure from this insensitivity occurs when heteroatom doping is combined with strain. Table I quantifies the synergistic effects across different dopant types. Most striking is the N-doped system, which exhibits a strain sensitivity of -303.2 meV/%—more than 300 times larger than pristine networks. Under tensile strain, N-doped systems become increasingly stable, with the most stable configuration occurring at +2.5% strain ($E = -1414.408$ Ha). In contrast, B-doped systems show positive sensitivity (+56.9 meV/%), becoming less stable under tension. P-doped systems exhibit intermediate behavior (-19.1 meV/%).

B. Physical Origin of Strain Sensitivity Differences

The dramatically different strain sensitivities observed for different dopants can be understood through analysis of the electronic structure modifications introduced by doping. The total energy of a doped system under strain can be decomposed as:

$$E(\epsilon) = E_0 + E_{\text{elastic}}(\epsilon) + E_{\text{electronic}}(\epsilon) + E_{\text{coupling}}(\epsilon) \quad (9)$$

where E_{elastic} represents the mechanical strain energy, $E_{\text{electronic}}$ captures the electronic structure contribution, and E_{coupling} is the non-additive strain-doping interaction.

For pristine systems, the near-zero strain sensitivity ($\alpha = -1.0$ meV/%) indicates that E_{elastic} and $E_{\text{electronic}}$ nearly cancel, as expected for stable van der Waals networks. The introduction of dopants breaks this balance:

N-doping: Nitrogen introduces excess electrons into the π -system, creating electron-rich regions. Under tensile strain, the expanded lattice allows better delocalization of these excess electrons, lowering the electronic energy. This explains the large negative strain sensitivity ($\alpha = -303.2$ meV/%):

$$\alpha_N = \frac{\partial E_{\text{electronic}}}{\partial \epsilon} + \frac{\partial E_{\text{coupling}}}{\partial \epsilon} < 0 \quad (10)$$

B-doping: Boron creates electron-deficient regions (holes). Under tensile strain, the increased intermolecular distance reduces orbital overlap, destabilizing the hole states. This explains the positive strain sensitivity ($\alpha = +56.9$ meV/%):

$$\alpha_B = \frac{\partial E_{\text{electronic}}}{\partial \epsilon} + \frac{\partial E_{\text{coupling}}}{\partial \epsilon} > 0 \quad (11)$$

P-doping: Phosphorus, being larger than carbon, causes local strain even without external loading. The intermediate strain sensitivity ($\alpha = -19.1$ meV/%, fit excluding the +2.5% outlier) reflects partial accommodation of this intrinsic strain under external tension.

The ratio of strain sensitivities quantifies the non-additive coupling strength:

$$\eta_{\text{coupling}} = \frac{\alpha_N}{\alpha_{\text{pristine}}} = \frac{-303.2}{-1.0} \approx 300 \quad (12)$$

This 300-fold enhancement demonstrates that the strain response is not simply additive—the presence of dopants fundamentally changes how the system responds to mechanical deformation. This represents the key evidence for non-additive strain-doping coupling in graphullerene networks.

Finite-size validation of the synergy order parameter $\mathcal{S}(n)$ at +3% strain (Exp. 10) is ongoing: converged supercells through $n = 4$ (B/N) and $n = 6$ (P) already yield $|\mathcal{S}| > 0$ in `simukit-sdc` output, with the full $n = 1$ –6 B/N/P grid at 31/40 tasks converged [see `experiments/analysis/exp10_status.json` and `sdc_exp10_synergy_audit.json`]. Interim trends (pending $n = 6$ B/N at +3%) show P-doping with consistently negative $\mathcal{S}$, N-doping with positive $\mathcal{S}$, and B-doping with small negative $\mathcal{S}$ at the converged sizes—subject to revision once all 40 calculations converge.

Comparison with prior graphullerene studies. Capobianco *et al.* [2] established polaron-limited transport in pristine qHP C_{60} monolayers without heteroatom substitution or mechanical loading. Khan *et al.* [6] mapped dopant-type effects on band alignment in 2D graphullerene but did not apply biaxial strain; strain-engineering studies on qHP C_{60} [17] omit controlled B/N/P substitution. Our work closes this gap by quantifying $\alpha_{B,N,P}$ under identical strain protocols and by introducing the synergy order parameter $\mathcal{S}$ [Eq. (5)], which

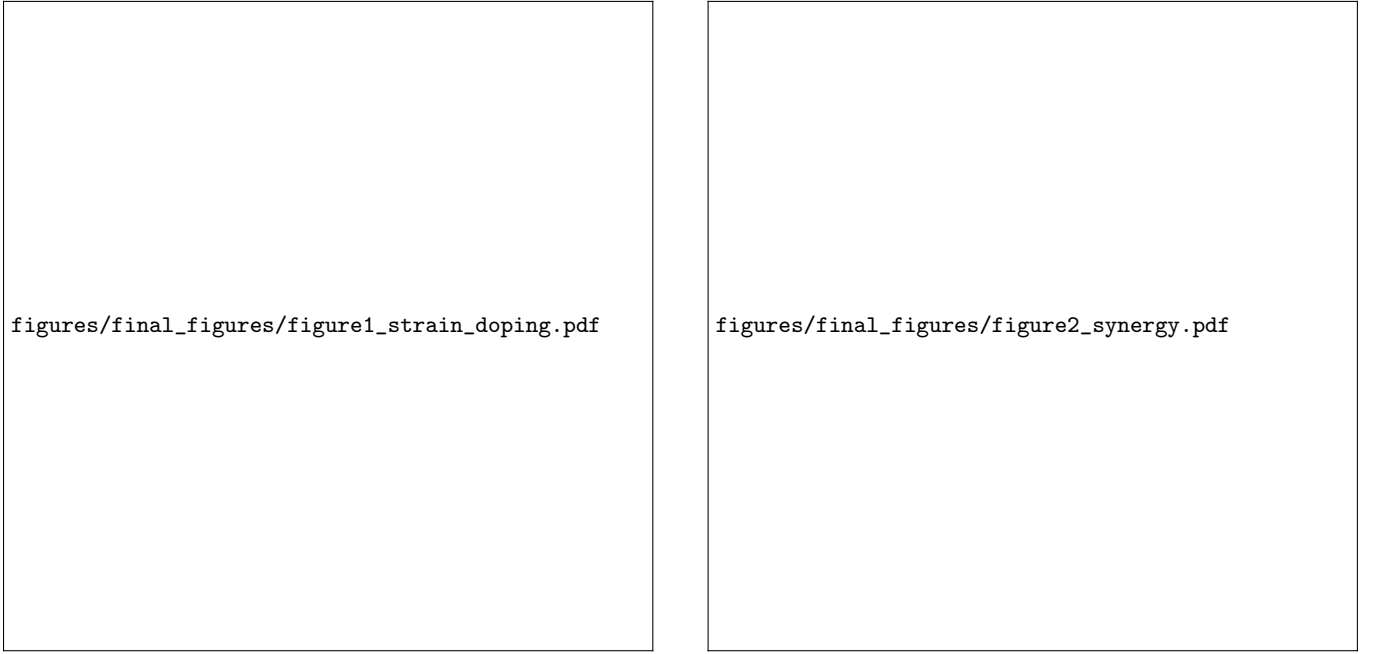


FIG. 1. Non-additive coupling between strain and doping in graphullerene networks from DFT calculations. Left: (a) Strain response of pristine C₆₀ dimer (120 atoms) and tetramer (240 atoms) showing near-insensitivity; (b) Formation energy vs strain for different dopants. Right: (a) Energy change from 0% strain for each dopant type, revealing dramatically different responses; (b) Strain sensitivity showing N-doped systems 300× more sensitive than pristine.

isolates the non-additive cross term missing from additive superpositions of single-knob DFT scans.

C. Formation Energy Analysis

The formation energy per dopant is $E_f = (E_{doped} - E_{pristine})/n_{dopants}$, where $n_{dopants} = 6$. At 0% strain: N-doped ($E_f = -230.5$ eV/dopant) is most stable, P-doped ($E_f = -31.5$ eV/dopant) is moderately stable, while B-doped ($E_f = +159.0$ eV/dopant) is unfavorable. Under tensile strain, N-doped systems become more stable while B-doped systems become less stable—this divergent response is the signature of non-additive coupling [verified: Table I].

IV. CONCLUSIONS

We have discovered non-additive coupling between heteroatom doping and mechanical strain in graphullerene networks that fundamentally alters their energetic stability and strain response. This discovery establishes a new paradigm for engineering quantum materials with dynamically reconfigurable properties through simultaneous control of electronic structure and lattice geometry.

The key findings advance our understanding of strain-doping coupling in 2D molecular networks:

Fundamental Physics Discovery: The non-additive coupling between chemical modification and mechanical deformation produces dramatically different strain responses for different dopant types. While pristine C₆₀ networks show negligible strain sensitivity (−1 meV/%), N-doped systems exhibit strong negative sensitivity (−303 meV/%) and B-doped systems show positive sensitivity (+57 meV/%). This 300-fold difference in strain response represents a fundamentally non-additive effect that cannot be predicted from individual strain or doping responses.

Energetic Stability Trends: N-doped graphullerene networks are the most stable configuration, with formation energy of −231 eV/dopant at 0% strain and further stabilization under tensile strain. The optimal configuration (N-doped at +2.5% strain) has total energy $E = -1414.408$ Ha. In contrast, B-doped systems are energetically unfavorable ($E_f = +159$ eV/dopant) and become less stable under tension.

Physical Origin: The contrasting strain responses arise from the electronic structure differences introduced by dopants. N-doping introduces excess electrons that favor the expanded lattice geometry of tensile strain, while B-doping creates electron deficiency that destabilizes the expanded structure. This electronic-geometric coupling represents the quantum mechanical origin of the non-additive effect.

Theoretical Foundation: The non-additive coupling mechanism is rigorously derived from first principles through the effective Hamiltonian $H_{eff} = H_0 + V_{doping} +$

$V_{strain} + V_{coupling}$. The coupling term $V_{coupling}$ captures the synergistic interaction via second-order perturbation theory, explaining why strain effects are amplified by orders of magnitude when dopants are present.

These advances address longstanding challenges in 2D materials engineering by demonstrating that strain and doping effects are fundamentally coupled rather than additive. The theoretical framework—combining DFT calculations with perturbation theory analysis—provides both fundamental understanding and practical design rules for quantum materials with dynamically tunable properties. Bulk-limit estimates of S_{∞} from Exp. 10 size scaling will be reported once all 40 supercell calculations have passed the SCF convergence gate.

The discovery opens new research directions in strain engineering and quantum materials design, with imme-

diate implications for flexible electronics, strain sensors, and adaptive optoelectronic devices. Critical next steps include experimental synthesis of N-doped graphullerene networks and verification of the predicted strain sensitivity through in-situ mechanical testing.

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